

New Noncontacting Inductive Analog Proximity and Inductive Linear Displacement Sensors for Industrial Automation

Sorin Fericean, *Senior Member, IEEE*, and Reinhard Droxler

Abstract—Noncontacting inductive sensors are applicable on a large scale for position detection or travel measurement in industrial applications. Reasons for such broad acceptance in many sectors of industry are noncontact and wear-free sensing of the target (any metal object), reliability and robustness, resistance to fouling, water tightness and compact size.

The present work is intended to be a systematic, complete, and consistent presentation of the technological innovations, recent implementations and current trends regarding the analog distance and travel sensing offered by noncontacting inductive sensors for industrial applications. It starts with the fundamentals of inductive sensing and presents the physical basics gained by modern analytic and simulation methods, as well as high-level integrated circuits for inductive sensors. The following sections deal with present-day inductive analog proximity sensors and with the distinctive technological innovation offered by the new inductive linear displacement sensors and with miniaturization results achieved through consistent integration.

Index Terms—Analog, application specific integrated circuits (ASICs), coil, displacement, distance, inductive, linear, proximity, sensor, travel.

I. INTRODUCTION

IN MODERN industrial production processes, the position of moving metal objects is ideally detected without mechanical contact and wear-free. The requirements for reliability, robustness, resistance to fouling, water tightness, etc., are severe. Electrical data such as temperature and supply voltage ranges, repeatability, accuracy, protection functions, and EMC requirements are firmly defined in standards [1]. These requirements are completely fulfilled by the *noncontacting inductive sensors*. These convert the distance between a metallic target and the sensor active face or the travel of the target, which performs a translatory movement in parallel to the sensor active face, into an electrical analog or binary signal [2].

According to the general IEEE-definition of physical sensors (Fig. 1) [3], the *primary transducer* of the inductive sensor is a coil.

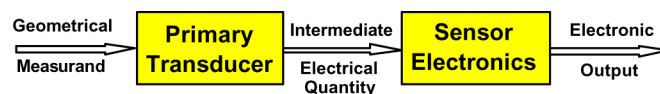


Fig. 1. Block diagram of a physical sensor.

The conversion of the geometrical measurand (distance or target travel, respectively) into an initial electrical quantity is a result of the coil excitation by the *sensor electronics*. The interference between the electromagnetic field generated by the coil and the target leads to modifications of the primary transducer electrical parameters. The sensor electronics evaluates these parameters and provides an electronic output signal, which is directly proportional to the geometrical measurand for analog sensor versions.

A popular version of the noncontacting inductive sensors is represented by the *inductive proximity* sensors, which have a respectable history of more than thirty years. After a long phase, characterized by modest parameters obtained through discrete electronics, a revival of the improvement activities was evident at the beginning of the 1990s, initiated by the large-scale integration of the electronic components. The fundamentals of these sensors, the technological progress, and the achieved status in year 2005 have been presented in [4] and [5].

In parallel to a great number of positive features, the inductive proximity sensor presents some restrictions as a consequence of physically given factors. The distance measuring ranges are small and strongly dependent on the sensor size. In addition, the sensors are not by themselves able to measure translatory movements of the target.

To overcome this obstacle, ingenious users usually complete the system with a metal incline, which is rigidly coupled with the object. The large displacement is converted mechanically into a small change of the distance between the incline and the active face of the inductive proximity sensor. This method enables significant increases in the sensing range and is the basis for the use of the inductive analog proximity sensor for clamping distance monitoring on tool spindles (Fig. 2) and work-piece clamping cylinders. In the former case, the task is to monitor whether the clamping device is open or closed with or without the tool, and in the latter case whether the work piece is correctly clamped. The function of the incline is performed by a cone located on the device shaft.

Unfortunately, this method has a negative influence on the measurement parameters. As a result, the positive features of the inductive proximity sensor, namely, linearity, accuracy, repeatability, etc., will be compromised by the incline's conversion ratio (usual values: 5–10).

Manuscript received May 8, 2007; revised July 23, 2007; accepted July 26, 2007. Expanded from a paper presented at the Sensors 2006 Conference. The associate editor coordinating the review of this paper and approving it for publication was Prof. Gerald Gerlach.

The authors are with the Corporate Innovation Management, Balluff GmbH, 73765 Neuhausen, Germany (e-mail: sorin.fericean@balluff.de; reinhard.droxler@balluff.de).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/JSEN.2007.908232

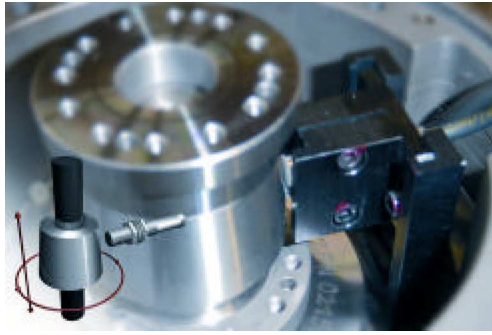


Fig. 2. Photo of inductive proximity sensor use for clamping distance monitoring in a milling machine and symbolic pictogram representation (left corner) of the conversion: axial travel of a rotating cone into distance to a cylindrical sensor [6].

A better approach to converting the inductive proximity sensor into a displacement sensor is the implementation of the travel information into the sensing element. The magneto-inductive sensor described in [7] is a displacement sensor which detects the translatory movement of an object with the aid of a permanent magnet (auxiliary target).

The main topic of this paper is to show how we could definitively eliminate both impediments resulting in the usage of the intermediate mechanical conversion and/or of the auxiliary magnet target with simultaneous preserving of the high qualities specific for the inductive proximity sensor. The final result is a new noncontacting inductive sensor, namely, the *inductive linear displacement sensor*.

II. FUNDAMENTALS OF THE INDUCTIVE LINEAR DISPLACEMENT SENSOR

A. Physical Basics

In principle, the noncontacting inductive sensors realize a continuous monitoring of the electromagnetic interference between a metallic target and the sensor primary transducer. The evaluation of the distance s or travel x is realized by continuous analysis of the impedance $\underline{Z}$ of the primary transducer, with its real and imaginary parts $\Re(\underline{Z})$ and $\Im(\underline{Z})$, respectively

$$\underline{Z} = \Re(\underline{Z}) + j\Im(\underline{Z}). \quad (1)$$

The dependences of these parts on electrical and mechanical parameters of the primary transducer are very complex and cannot be expressed by analytical equations. Logical initial reflections allow primary transducer designs for which these parts show univocal dependencies on the working frequency f and the variable s or x (written $s|x$)

$$\begin{aligned} \Re(\underline{Z}) &= F_R(\dots, s|x, f, \dots) \\ \Im(\underline{Z}) &= F_I(\dots, s|x, f, \dots). \end{aligned} \quad (2)$$

Based on our experience in the design and production and on our intensive research with focus on simulations and high performance, we can describe the physical basics of the noncontacting inductive sensors making a segmentation of the working frequency range in three intervals.

- 1) For comparatively high frequencies ($f = 1$ to 5 MHz), the magnetic properties of ferromagnetic targets are diminished radically, so that the behavior of the primary transducer relative to the target can be reduced to the eddy-current effect [8]. A particular family of inductive proximity sensors with coreless coils, which does not make a distinction between ferromagnetic and nonferromagnetic targets, is based on this fact.
- 2) In the characteristic frequency range of noncontacting inductive sensors ($f = 50$ kHz to 1 MHz), the behavior of the primary transducer depends very strongly on the target properties (ferromagnetic or nonferromagnetic). The magnetic field generated by the transducer-coil induces eddy-currents in the target. The electrical losses in the target depend on the strength of the magnetic field, the material properties of the target, and on the travel x . These result in a change in the electrical parameters of the primary transducer-coil. This does not, however, fully describe the function principle of the sensors, since most of them possess a transducer-coil with a core. Together with the ferromagnetic target, the core forms a magnetic circuit whose magnetic reluctance depends on the distance. The varying of this resistance causes a change in the coil inductivity.
- 3) For low frequencies the eddy-current effect becomes negligible and the behavior is highly dependent on the target magnetic properties. This frequency range is less applicable for the noncontacting inductive sensors and will only be used for special versions like “ferrous or nonferrous” inductive proximity sensors, which provide a selective detection of the target. At the same time, the range is specific for inductive sensors which require a permanent mechanical contact between the object to be detected and sensor, like variable reluctance sensors, LVDTs, PLCDs, etc., [2], [9].

B. Evaluation Procedures With Single-Coil Transducer

The most suitable and cost-effective method for the evaluation of the parameters of the primary transducer consisting of a single coil is the continuous measuring of its q-factor Q_L , which is a univocal function versus s or x

$$Q_L(s|x) = \frac{\Im(\underline{Z})}{\Re(\underline{Z})}. \quad (3)$$

Using the applied electromagnetic theory and/or CAD simulation tools [10]—which basically solve Maxwell’s equations for the specific primary transducer structure—the q-factor can be calculated in terms of field quantities with very good accuracy (below 1%) with the expression

$$Q_L = \frac{\omega \cdot \sum_{i=1}^n \mu_i \iiint_i H_i(p) \bullet H_i^*(p) dv}{\sum_{i=1}^n \frac{1}{\sigma_i} \iiint_i J_i(p) \bullet J_i^*(p) dv} \quad (4)$$

where the vectors $\mathbf{H}$ and $\mathbf{J}$ represent the magnetic field strength and conduction current density, respectively, as functions of the position vector $\mathbf{p}$. The scalars μ and σ are the magnetic permeability and electrical conductivity; ω is the angular frequency. The index i represents the current number of the n system components. This method permits an accurate estimation of the transducer performance [5].

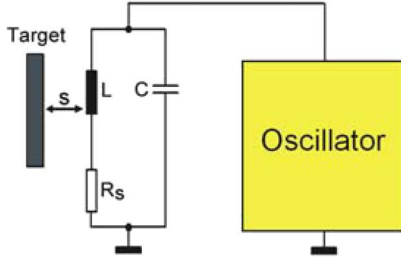


Fig. 3. Excitation of the sensor primary transducer by an oscillator with negative resistance. The primary transducer coil is represented by serial components: inductivity L and total loss resistance R_S .

Using the Jordan equivalent circuit of a coil, which consists of a serial connection of the inductivity L and total loss resistance R_S [4], and the angular frequency ω the q-factor Q_L can be particularized to

$$Q_L(s|x) = \frac{\omega \cdot L}{R_S}. \quad (5)$$

To implement measurement of the q-factor in the noncontacting inductive sensors the primary transducer-coil and a low loss capacitor C are connected to a parallel resonant circuit and excited at the resonant frequency (Fig. 3). We use the proven oscillator with negative resistance [11], [12], which performs a direct conversion of the q-factor into an electrical signal and allows a two-wire connection.

Besides the advantages resulting in only two connection wires to the coil, adjustment flexibility, greater possibilities for linearization and efficient temperature compensation [13], the oscillator with negative resistance can be integrated in an ASIC using various technologies from bipolar to BiCMOS. The oscillator can be very conveniently adjusted by a means of a trimming resistor. The resonance frequency f_R depends only on the primary transducer-coil parameters L and R_S and on the capacity C (Fig. 3) and can be considered independent on the travel x

$$f_R = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R_S^2}{4L^2}}. \quad (6)$$

Due to the nearly ideal working conditions (very high input impedance of the oscillator, negligible losses in the capacitor), the q-factor Q of the resonant circuit is reciprocal to the Q_L -factor of the coil and, consequently, only depends on the primary transducer-coil parameters L and R_S , respectively, on the working frequency f_R

$$Q = 1/Q_L. \quad (7)$$

In addition, the impedance of the parallel resonant circuit working at its resonance frequency f_R is a real-value R_P , which is directly proportional to the Q_L value

$$R_P = \frac{Q_L}{2\pi \cdot f_R \cdot C}. \quad (8)$$

Finally, the sinus wave $v_{LC}(t)$ across the resonant circuit supplied with constant current I_E by the oscillator has an amplitude

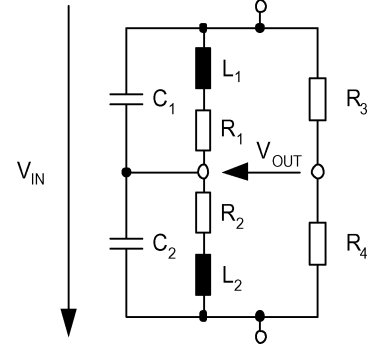


Fig. 4. Double resonant oscillator bridge. The primary transducer coils are represented by serial components: inductivity L_i and total loss resistance R_i [6].

directly proportional to R_P and, therefore, exactly reproduces the dependence $Q_L(s|x)$

$$v_{LC}(t) = R_P \cdot I_E \cdot \sin(2\pi \cdot f_R \cdot t). \quad (9)$$

In conclusion, the measurement of the sinus wave amplitude offers precise and feasible direct information about the q-factor of the primary transducer and thereby indirect information about the travel x .

C. Evaluation Procedures With Double-Coil Transducer

The evaluation of the quality factor using a single coil has several disadvantages. First, the q-factor is strongly temperature-dependent and difficult to compensate. Second, the linearity error of the travel measurement is affected by the temperature dependence of the q-factor.

For this reason, we elaborated an advanced evaluation procedure for high accuracy applications based on a double-coil transducer. There is a proven balancing technique which uses a double-coil and exploits the properties of a detuned oscillator bridge. The primary transducer consists of two identical coils which are reversely influenced by the target movement. The coils are embedded in an original new double resonant circuit (Fig. 4). The right side of the bridge is purely resistive. The left side of the oscillating bridge comprises two separate resonant circuits with the coils (represented by their inductivities L_1, L_2 and series resistances R_1, R_2) and the corresponding capacitors.

If L_1, L_2 and C_1, C_2 have the same values, respectively, the resonant frequencies are approximately equal. In addition, the detuning of the resonant frequency as a result of a target movement can be ignored. Assuming that the oscillator bridge is excited with this resonant frequency, the resonant circuits can be replaced by parallel resistances R_{P1}, R_{P2} (8) to evaluate the transfer function

$$H_0 = \frac{V_{OUT}}{V_{IN}} = \frac{1}{2} \frac{R_{P1} - R_{P2}}{R_{P1} + R_{P2}} - \frac{1}{2} \frac{R_3 - R_4}{R_3 + R_4}. \quad (10)$$

With the quality factors Q_{L1}, Q_{L2} , and the angular resonant frequency ω_R , the parallel resistances result from (8)

$$R_{P1} = Q_{L1} \cdot \omega_R L_1, \quad R_{P2} = Q_{L2} \cdot \omega_R L_2. \quad (11)$$

Using (10) and (11), the transfer function can be written as

$$H_0 = \frac{V_{OUT}}{V_{IN}} = \frac{1}{2} \frac{Q_{L1} - Q_{L2}}{Q_{L1} + Q_{L2}} - \frac{1}{2} \frac{R_3 - R_4}{R_3 + R_4} \quad (12)$$

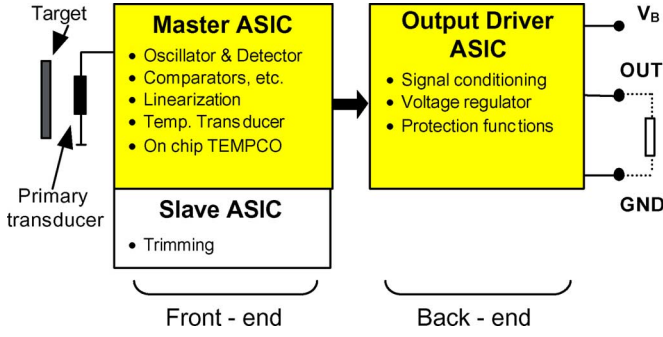


Fig. 5. General block diagram for the base circuit of the noncontacting inductive sensors. The slave ASIC for trimming is optional.

which—apart from a constant term—only depends on the ratio of the two q -factors Q_{L1} and Q_{L2} .

The direct evaluation of the transfer function as a ratio of two voltages is either cumbersome or imprecise. A very simple approach to precise evaluation is to keep the input quantity V_{IN} constant by using feedback. Then, the output quantity V_{OUT} is proportional to the transfer function

$$V_{OUT} = V_{IN} \cdot H_0(Q_{L1}, Q_{L2}). \quad (13)$$

III. STRUCTURES AND OPERATION OF THE SENSOR ELECTRONICS FOR NONCONTACTING INDUCTIVE SENSORS

The sensor electronics of a noncontacting inductive sensor can be basically divided into two parts. The front-end includes the excitation of the primary transducer and the signal processing to convert the geometrical measurand into an intermediate electrical quantity. Successively, this value is provided to the back-end section which performs the final signal conditioning in order to generate output signals, which conform to the standards. At the same time, the back-end acts as interface between the sensor and the application, offering the standard protection functions [1].

Sensor realizations with discrete components belong to history and can be found at present only in low-cost, low-performance versions. In the last years, we developed two generations of application specific integrated circuits (ASIC) for our noncontacting inductive sensors. The modular multichip structure represents the best compromise between performance, cost, and flexibility (Fig. 5). The grounded-base circuit comprises only two ASICs and is used for Balluff noncontacting inductive sensors.

The master ASIC comprises the stages for the entire front-end operation and is integrated with a low-voltage, high-frequency bipolar process. The fine monolithic structure allows integration of more than 700 transistors on a silicon area of less than 4.5 mm^2 [14].

The output driver ASIC for final output signal generation and sensor protections (short circuit, reverse polarity, and open-wire protection, as well as clamping of the kickback pulses due to switching inductive loads) is realized by means of a medium-voltage process. This stage makes the sensor suitable for supply voltages up to 30 VDC and delivers a regulated supply voltage for the front-end.

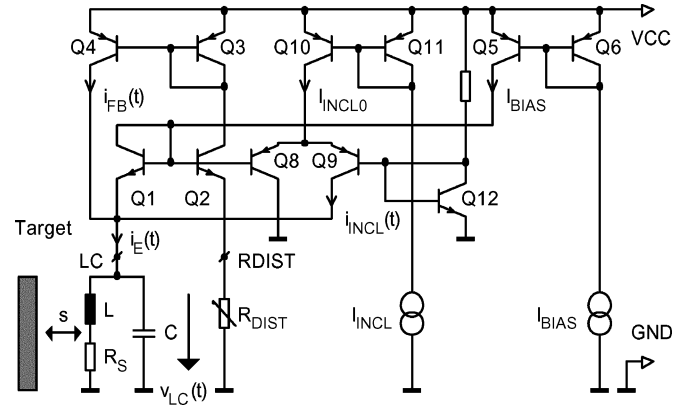


Fig. 6. Typical schematic of an oscillator with negative resistance in bipolar version and with linearization circuit [15].

IV. SIGNIFICANT IMPROVEMENTS IN THE INDUCTIVE ANALOG PROXIMITY SENSOR, PRESENT-DAY VERSIONS AND FEATURES

The inductive analog proximity sensors are approaching sensors, which detect the movement of the metal target in a direction perpendicular to their active face and provide output signals (voltage or current) directly proportional to the distance between target and sensor. They basically are inductive proximity sensors and firstly have to meet the general requirements for this sensor family [1]. Secondly, they also have to meet demands which are specific for measurement systems, namely, accuracy and repeatability, resolution and linearity.

For inductive analog proximity sensors the specific difficulty can be found in the oscillator. In contrast to binary sensors, the oscillation amplitude must have a linear dependence on the distance [4]. The task has been solved by an adequate optimization of the primary transducer and by the elaboration of an original oscillator circuit. The U.S. patent [15] describes the details of the ring oscillator's linearization mechanism and gives more concrete implementation solutions. The basic idea is the generation of a primary transducer-coil excitation current dependent on the distance s , which has to compensate for the quasi-exponential character of the dependence of $Q_L(s)$ [5].

A simple and easy-to-integrate oscillator with linearization is illustrated in Fig. 6 [15]. The feedback current is an addition of two partial currents and can be adjusted in order to compensate for the nonlinear variation $Q_L(s)$. The partial current $i_{FB}(t)$ generated by the ring oscillator has an amplitude directly proportional to the voltage $v_{LC}(t)$ over the LC-resonant circuit and is adjustable by an external resistor R_{DIST} . A second nonlinear partial current $i_{INCL}(t)$ for the inclination control of the oscillator characteristic is provided by a differential amplifier and biased by a DC current I_{INCL0}

$$i_{INCL}(t) = \frac{I_{INCL0}}{2} \cdot \left[1 + \tanh \frac{v_{LC}(t)}{2V_{th}} \right] \quad (14)$$

where V_{th} is the thermal voltage. The matching between the partial currents provides a linearized oscillator characteristic, amplitude of the signal $v_{LC}(t)$ versus distance s .

For analog sensors temperature dependence is an important challenge, since the entire characteristic of the sensor must be independent of the ambient temperature. Temperature compensation takes place in two steps [13]. The initial compensation is achieved by optimum design of the oscillator working frequency. The second step consists of an active compensation,

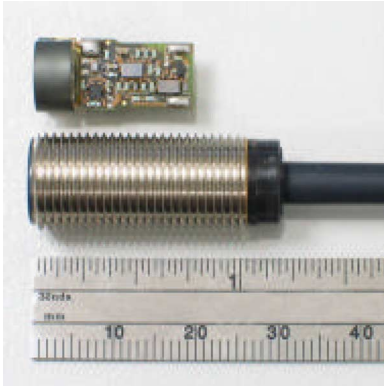


Fig. 7. Photo of a high miniaturized and high-performance inductive analog proximity sensor (subassembly with coil and sensor with metal housing).

which is performed by a temperature compensation circuit integrated into the master ASIC. The temperature drift of the primary transducer is determined in the development phase for each sensor type and approximated in the temperature compensation circuit. During operation, the excitation current of the sensing element is modified to compensate the temperature drift.

Precise setting with low sample variations is achieved by a *teach-in procedure* after the sensors have been fully assembled. The calibration is done using a standard target which is positioned at the upper limit of the distance range. Communication between the sensor and the calibration device takes place over the supply lines of the sensor (no additional connections required). Once the calibration commences, the oscillator amplitude is changed by means of a programmable resistor network integrated into the slave ASIC (Fig. 5), until the sensor output signal reaches the required upper limit. The resistance value is stored in an EEPROM in the slave ASIC. When the calibration is finished, the communication interface is deactivated so that changes due to parasitic ambient conditions are not possible. The teach-in compensates for all parameters which can affect the output signal—including the properties of the potting material or position tolerance of the sensing element in the sensor housing—for each individual sensor.

The inductive analog proximity sensors (Fig. 7) described here have either a voltage output of 0...10 V or a current output of 0...20 mA, respectively, 4...20 mA [16]. The sensors can be operated over a supply voltage range of 15–30 V (voltage output) or 10–30 V (current output). Typical data are a linearity error of less than $\pm 3\%$ and a resolution limit of 0.1% (full scale).

Experience shows that the spectrum of possible applications is constantly being enlarged. Examples for the variety of industrial applications include: thickness measurement of dielectric foils, layers; detection of nonhomogenous zones of level metal faces (gap, slot, etc.); runout inspection; counting and monitoring tasks; detecting the orientation of different parts in work piece inspection.

V. THE DISTINCTIVE TECHNOLOGICAL INNOVATION REPRESENTED BY THE INDUCTIVE LINEAR DISPLACEMENT SENSOR

A. Measuring Principle

The new inductive linear displacement sensor operates without contact and detects the position of a slender metallic

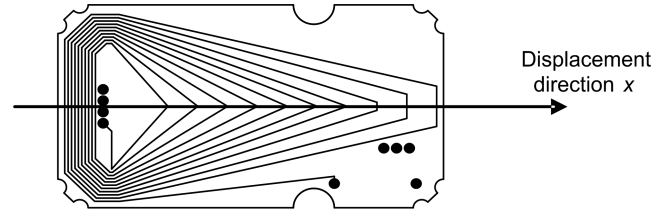


Fig. 8. Drawing of the single-coil primary transducer of the inductive linear displacement sensor (dimensions 30 mm $\times$ 10 mm) [6].

target which performs a translatory movement in parallel to the sensor active face. The target acts locally on the sensor primary transducer and influences its q -factor which is processed by the sensor electronics. This influence is proportional to the target projection on the primary transducer. In consequence of that, if the geometry of the primary transducer varies over the measuring path—implementation of the travel information—the result is a univocal correlation between q -factor and displacement x .

B. Primary Transducer

The radical change in the structure of the primary transducer of the inductive linear displacement sensor in contrast to inductive proximity sensors is the reshaping of the classical wire-wound coil with round core into a planar structured coil on dielectric substrate (Fig. 8) with flat core (ferrite foil) on its rear surface.

The target acts on the planar sensor coil excited by the sensor electronics with a high-frequency current and causes a position-dependent change in the coil q -factor [5], which is continuously measured by the sensor electronics. The function Q_L versus displacement x can be optimized by geometrical changes into the shape of the triangles, for example, in

$$l(m) = a \cdot m^c + b \quad (15)$$

where $l(m)$ is the length of the triangle having the serial number m . The constants a and b , as well as the exponent c are geometrical design values which depend on coil size and features. The shape in Fig. 8 corresponds to an arithmetical progression ($c = 1$).

The planar coil allows for a very convenient implementation of the travel information. Fig. 8 illustrates our final topology with ten windings shaped as interleaved isosceles triangles. The function Q_L versus displacement x can be optimized by geometrical changes into the shape of the triangles. We use an eight-layer planar structure with eight such coils connected in series. The high mutual coupling between the layers results in a total inductivity of around 100 μH and high q -factor values in respect of planar coils.

Fig. 9 shows the dependence Q_L versus displacement x for the above described coil obtained by the measuring of the electrical parameters L and R_S and applying (5). The characteristic shows a monotonic range from -17 to 0 mm, in which a sensing operation can be implemented.

The analytical approach, consisting of CAD simulations [10] with the corresponding coil model (Fig. 10) and applying of expression (4), provided very similar results.

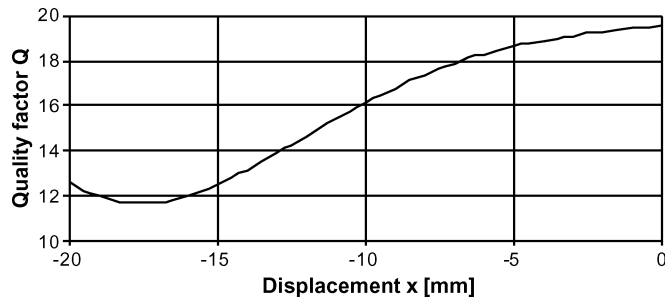


Fig. 9. Quality factor characteristic of the single-coil primary transducer illustrated in Fig. 8 [6].

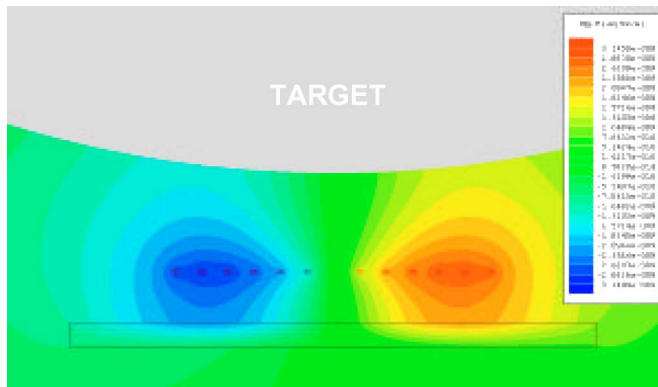


Fig. 10. Magnetic field of the coil illustrated in Fig. 8 with round target. Simplified simulation model (only one layer) and Cartesian ZY -representation for a cross section in the middle of the coil and perpendicular to the x -direction. Visible are six windings, the ferrite foil, and a segment of the round target [6].

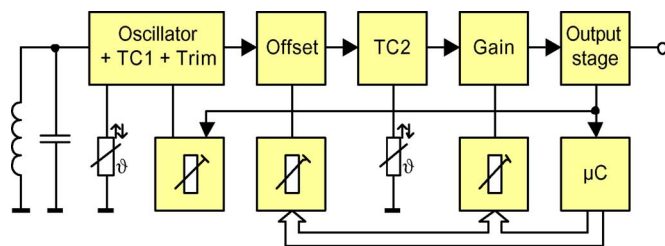


Fig. 11. Block diagram of the inductive linear displacement sensor [6].

C. Operation of the Sensor Electronics

For this sensor, the electronics can be basically divided into three parts (Fig. 11). The front-end is very similar to the version of the inductive analog proximity sensor (Fig. 5). Besides the oscillator, which provides the excitation and conversion functions, this part includes the aforementioned temperature compensation (TC1), which is now employed to compensate for the temperature drift of the starting point of the distance/output characteristic (geometrical correspondent is the left coil side in Fig. 8) and the base-trimming circuit realized by the slave ASIC.

The interface performs further signal conditioning with a second temperature compensation stage (TC2), which compensates for the temperature drift at the end point of the measuring range, and two adjustable amplifiers for offset and gain adjustment. These amplifiers are digitally controlled by means of digital potentiometers and a microcontroller, whereas the signal path still remains pure analog. The third part is an output stage similar to Fig. 5.

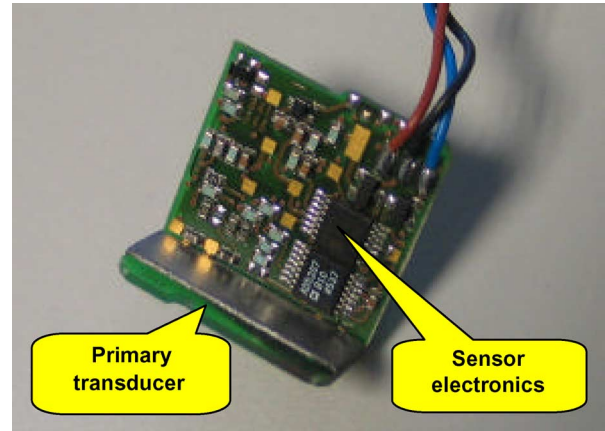


Fig. 12. Subassembly of an inductive linear displacement sensor. The carrier PCB for the sensor electronics is mounted perpendicular to the primary transducer consisting of a PCB coil (Fig. 8) and ferrite foil.

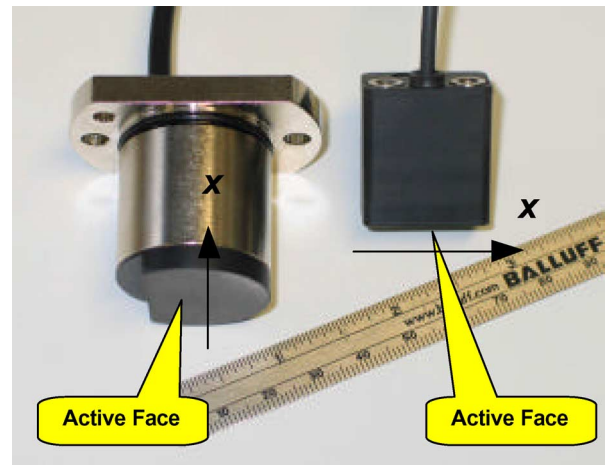


Fig. 13. Robust inductive linear displacement sensor version in a metal housing with a mounting flange and a plastic cap as active face (left). Miniaturized version in a plastic housing without mounting flange (right) [6].

D. Realization, Feature and Specific Applications

The first inductive linear displacement sensor version presented above has already been put into practice. It reached the industrialization phase and passed field tests at reference customers. The focused displacement range specified for this first step was below 15 mm. Two compact mechanical designs have been followed. They have comparable electrical data and very similar internal structure (Fig. 12).

The embodiments strongly differ in the housing concept. There is a robust round version (30 mm diameter and 40 mm total length) in metal housing (Fig. 13 left), which has been designed for heavy duty usage, and a smaller cubical version (30 mm × 30 mm × 15 mm) in plastic housing (Fig. 13 right), which is suitable for miniature applications.

The sensors are fully potted, so that the requirements of IP67 and those regarding the immunity against mechanical shocks and vibrations are easily met within a large temperature range. This makes them useful even in harsh environmental conditions specific for industrial applications.

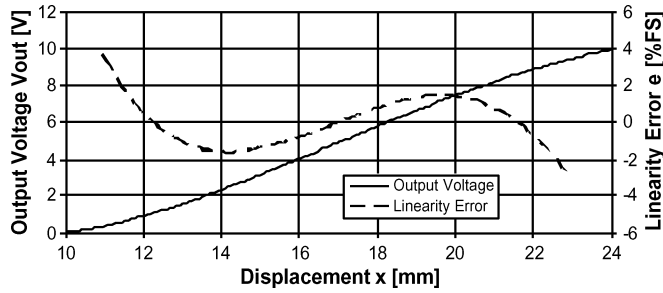


Fig. 14. Sensor output characteristic and linearity error curve measured with a steel reference target (travel range equal to 14 mm) [6].

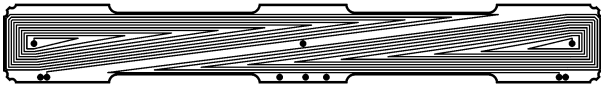


Fig. 15. Drawing of the double-coil primary transducer for the advanced inductive linear displacement sensor version.

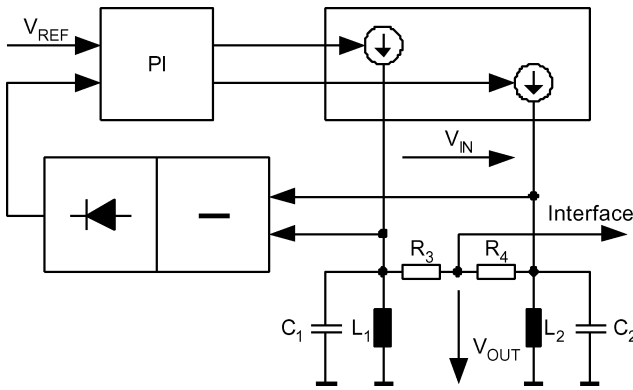


Fig. 16. Push-pull oscillator for the advanced inductive linear displacement sensor version [6].

Precise setting with low sample variation of the displacement/output characteristic is achieved by a calibrating procedure similar to the procedure described above (Section IV). Here, a *two-point teach-in procedure* is performed to adjust the offset and gain resistance networks (Fig. 11) in order to calibrate both points: start and end of the sensor characteristic.

The high accuracy achieved by this method can be seen in Fig. 14, which shows the measured distance/output characteristic and the nonlinearity (S-curve) in the range: 0 V @ 10 mm to 10 V @ 24 mm.

With linearity error of less than $\pm 3\%$ (for $x = 11$ to 23 mm) and resolution limit of 0.1% (full scale), the inductive linear displacement sensor is the solution offering the best cost/benefit ratio for many measuring or positioning tasks.

E. Advanced Implementation

This improved inductive linear displacement sensor version is based on travel evaluation with a double-coil primary transducer (Fig. 15), as described in Section II-C. The coils are embedded in the double resonant bridge (Fig. 4).

The resonant bridge can be excited using an external oscillator. It is however preferable to use the bridge itself as the resonant circuit of an oscillator. The oscillator shown schematically

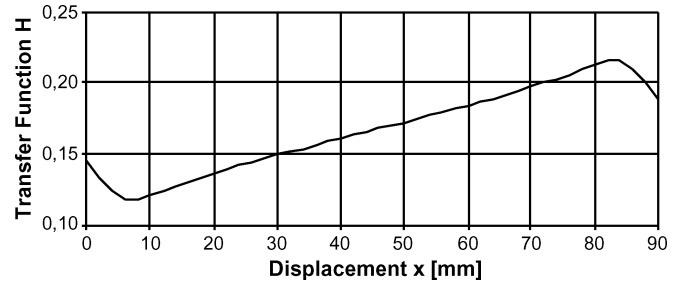


Fig. 17. Transfer function characteristic (measuring range 10–80 mm, primary transducer dimensions 92 mm $\times$ 12 mm) [6].

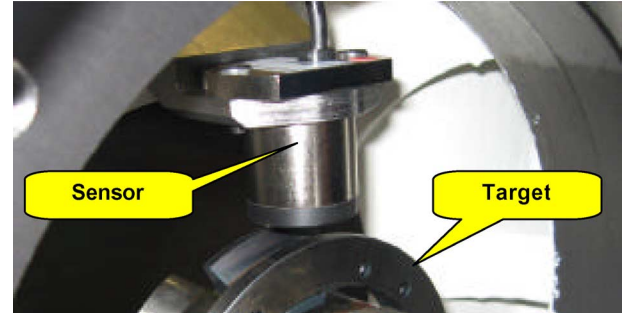


Fig. 18. inductive linear displacement sensor as innovative drop-in replacement of the inductive proximity sensor [6].

in Fig. 16 is implemented as a push-pull oscillator [6], [17]. This has the advantage that the output voltage V_{OUT} is referenced to ground. The oscillator amplitude V_{IN} is kept constant by means of a loop consisting of a subtractor, a peak detector and a PI controller. The oscillator provides an analog, time-continuous output signal V_{OUT} which only depends on the ratio of two q -factors (12), (13).

The transfer function corresponding to (12) and obtained with a prototype of the double-coil primary transducer is shown in Fig. 17.

The characteristic has a very large monotonic range of more than 70 mm—compared with 92 mm primary transducer length—with very good linearity. Common mode interferences will be rejected. Temperature drift and sample-to-sample tolerances are strongly reduced, because both coils are placed on a common PCB substrate.

Due to its flexibility, robustness, and compact construction, the inductive linear displacement sensors are ideal for numerous industrial applications and even open up areas for new implementations.

The standard application of the inductive linear displacement sensor is given by a machine part that performs a linear displacement. The sensor provides an output signal which is proportional to the displacement. The target can also make complex movements (Fig. 18). The target consists of a slender metal disc and performs a continuous rotation accomplished by a linear displacement.

VI. MINIATURIZATION THROUGH CONSISTENT INTEGRATION

Implementation of the high functionality in miniaturized housing was made possible through consistent integration of the electronic core components, and the use of the most

up-to-date assembly technologies. Highly integrated ASICs along with few *surface mounted technology* (SMT) components are assembled on *printed circuit boards* (PCBs) with up to ten layers and microvias.

For placing and connecting the ASIC devices in chip form (without housing) two modern *chip-on-board* (COB) technologies are used, wire bonding and *flip-chip-on-board* (FCOB) technology [18]. The highly modern FCOB method involves connecting the chip to the PCB directly by the aid of solder points (connection bumps). For this reason, the chip active surface is facing down; no protective covering is needed (Fig. 7).

Basically, this technology is a standard method in other electronics fields [18]. Our contribution [14] was to match the FCOB-technology to the FR4-material (*fiberglass-reinforced substrate*) used as carrier board for the electronics and, in this way, to adapt the FCOB technology to the thermal, mechanical and electrical high-stress conditions specific for the noncontacting inductive sensors. The qualification of the sensors confirmed the high reliability of our particular FCOB technology.

VII. CONCLUSION

When distances and positions up to 50 mm have to be detected, popular inductive analog proximity sensor remain the best alternative from all points of view. The evolution of the material, electronics, and assembling technologies considerably improved performance-cost ratio and maintained the inductive proximity sensors in the top position in the sensor world for industrial applications.

For displacements and positions from 50 to 200 mm, there has until recently been no contact-free alternative to the inherently wear-prone potentiometer. Users almost always had to compromise with respect to housing size, the expected useful life or the investment costs. The inductive linear displacement sensors from Balluff/Germany now close this gap.

ACKNOWLEDGMENT

The author would like to thank all co-workers in the Sensors Development Department for their cooperation and Balluff GmbH for providing the working conditions and permanent support.

REFERENCES

- [1] IEC 60947, Low-voltage switchgear and controlgear Parts 5-2 & 5-7: Control circuit devices and switching elements—Proximity switches and requirements for proximity sensors with analog output.
- [2] *Sensortechnik*. Berlin: Springer, 1998, H. R. Tränkler and E. Obermeier (Hrsg.).
- [3] R. Dorf, Ed., *The Electrical Engineering Handbook*. Piscataway, NJ: IEEE Press, 1993.
- [4] M. Jagiella and S. Fericean, "Miniaturized inductive sensors for industrial applications," in *Proc. 1st IEEE Conf. Sensors*, Orlando, FL, 2002, vol. II, pp. 771–778.

- [5] M. Jagiella, S. Fericean, A. Dorneich, and M. Eggmann, "Theoretical progress and recent realizations of miniaturized inductive sensors for automation," in *Proc. IEEE Conf. Sensors*, Irvine, CA, 2005, pp. 488–491.
- [6] M. Jagiella, S. Fericean, and R. Droxler, "New non-contacting linear displacement inductive sensors for industrial automation," in *Proc. IEEE Conf. Sensors*, Daegu, Korea, 2006, pp. 534–537.
- [7] M. Jagiella, S. Fericean, R. Droxler, and A. Dorneich, "New magneto-inductive sensing principle and its implementation in sensors for industrial applications," in *Proc. IEEE Conf. Sensors*, Vienna, Austria, 2004, pp. 1020–1023.
- [8] H. Loos, *Systemtechnik induktiver Weg- und Kraftaufnehmer*. Ehingen: Expert-Verlag, 1992.
- [9] R. Pallas-Areny and J. G. Webster, *Sensors and Signal Conditioning*, 2nd ed. New York: Wiley, 2001.
- [10] "Technical Manual: Maxwell-Simulator," Ansoft Corp., 2005.
- [11] A. A. Andronov, A. A. Vitt, and S. E. Khaikin, *Theory of Oscillators*. New York: Dover, 1987.
- [12] S. Fericean, M. Friedrich, M. Fritton, and T. Reider, "Moderne Wirbelstromsensoren—linear und temperaturstabil," *Elektronik*, no. 8, pp. 98–103, Apr. 2001, Germany.
- [13] M. Jagiella, S. Fericean, M. Friedrich, and A. Dorneich, "Mehrstufige Temperaturkompensation bei induktiven Sensoren," *Elektronik*, no. 16, pp. 62–66, Aug. 2003, Germany.
- [14] M. Jagiella, S. Fericean, and A. Dorneich, "Progress and recent realizations of miniaturized inductive proximity sensors for automation," *IEEE Sensor J.*, vol. 6, no. 6, pp. 1734–1741, Dec. 2006.
- [15] S. Fericean, M. Friedrich, and E. Gass, "Inductive sensor responsive to the distance to a conductive or magnetizable object," Balluff Patent 5,504,425, 1996.
- [16] Balluff Websites. [Online]. Available: www.balluff.de and www.balluff.com
- [17] R. Droxler, "Berührungslos arbeitender Näherungsschalter," Neuhausen, Germany Balluff Patent DE 19611810 C2, 2000.
- [18] H. J. Lau, *Flip Chip Technologies*. New York: McGraw-Hill, 1995.



Sorin Fericean (SM'05) was born in 1948. He received the degree in electronics from the Polytechnical University of Timisoara, Timisoara, Romania, in 1971.

In 1990, he joined Balluff GmbH, Neuhausen, Germany, as a Development Engineer and became the Leader of the Innovation Development Group in 1992. Since 2001, he has been the Head of the R&D Department, Business Unit for Sensors, Balluff GmbH. In this position, he worked on the basic development of new displacement and distance sensors and the implementation of medium- and large-scale integration of the corresponding electronics, reflected also by the present publication. In 2006, he assumed responsibility for the Corporate Innovation Management of the Balluff Group, with a core competence consisting of scouting, analysis, and evaluation, as well as basic design for future sensor technologies and operations.



Reinhard Droxler was born in 1959. He received the degree in electrical engineering from the Friderician University, Karlsruhe, Germany, in 1984.

In 1985, he joined Wandel & Goltermann GmbH, Eningen, Germany. Since 1994, he has been with the R&D Department, Business Unit for Sensors, Balluff GmbH, Neuhausen, Germany. In 2006, he changed to the Corporate Innovation Management of the Balluff Group, with a core competence consisting of scouting, analysis, and evaluation, as well as basic design for future sensor technologies and operations.

His research interests include the development and implementation of new sensor concepts, as well as analog and digital signal processing.